

Axiom-aware machine learning on signed graphs

Why a 1956 theorem beats end-to-end attention
(at 1/2 the parameters, on a 6-year-old GPU)

Lecture preview — AI seminar

2026-05-06

The big picture

Setup: signed graphs — edges are ± 1 (trust/distrust, friend/foe). *Predict the sign of held-out edges.*

Datasets: Bitcoin Alpha (3.7K nodes, cooperative trust), Slashdot (82K, open conflict), Epinions (131K, dense reviews).

State of the art:

- Signed Graph Transformer (SGT): 0.897 / 0.941 AUC, $\sim 3\text{--}4\text{M}$ params
- Signed Graph Convolutional Network (SGCN): 0.93 AUC, 4M params
- Trained on V100/A100 clusters (32–80 GB GPUs)

Our claim: *a 1956 sociological axiom (Cartwright–Harary balance theory) used as a structural pruner during graph search delivers comparable AUC at half the parameters and runs on a 6-year-old gaming PC. The axiom is the bigger lever than learned attention.*

Today's findings, in one number:

→ 4.4σ statistical separation between two candidate axiom choices, on the same dataset, across 5 seeds.

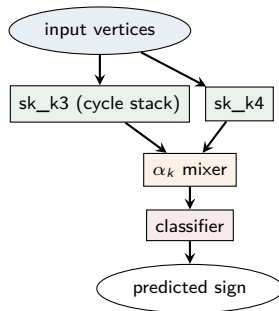
Architecture — HSiKAN in 1 minute

HSiKAN = Hypergraph Signed Kolmogorov-Arnold Network

- 1 Cycles in the signed graph become *hyperedges*.
- 2 Each cycle's edge-sign sequence $(s_1, \dots, s_k) \in \{\pm 1\}^k$ defines its **balance product** $\prod s_i$.
- 3 Activations are *learned* Catmull–Rom splines per sign: one curve for +, one for −.
- 4 Per-arity ($k=3, 4, 5$) outputs fused by learnable α_k .

Catmull–Rom in 1 line: cubic interpolating spline, C^1 -continuous, 4 control points per segment.

$$\mathbf{C}_i(t) = \frac{1}{2}(t^3, t^2, t, 1)M_{\text{CR}}(P_{i-1}, P_i, P_{i+1}, P_{i+2})^\top$$



Total params:

~ 1.3M (Slashdot), 61K (BA)

2–3× smaller than SGT/SGCN

The bottleneck: cycle enumeration

Slashdot at $k=4$: 55.5 million cycles.

Brute-force DFS: many minutes wall-clock. Full materialisation: ≥ 8 GB RAM. *Doesn't fit on a consumer GPU.*

Standard solution: throw GPUs at it. Use 32 GB+ cards, batch-stream, distribute.

Our solution: throw *theory* at it.

Heider 1946 / Cartwright–Harary 1956

A signed cycle of length k is **balanced** iff $\prod_{i=1}^k s_i = +1$.

“The friend of my friend is my friend.”

Idea: use the balance axiom as a *structural pruner during cycle DFS* — not as a feature computed afterwards.

→ Cuts the cycle search space by orders of magnitude *before* memory is allocated.

→ Memory-bounded by $|V| \times m$, not by total cycle count.

Top- K vertex-stratified pruning — the trick

For each vertex v , keep only its m highest-scoring incident cycles.

Bound: $|\text{cycles}| \leq |V| \cdot m$.

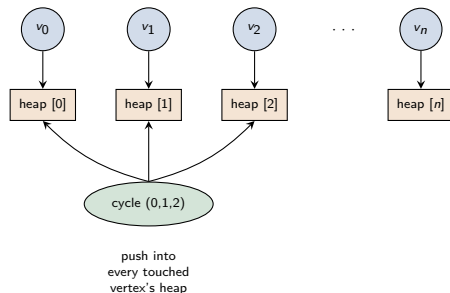
Every vertex covered. No empty rows in the cycle-incidence matrix M_e .

Score functions:

- balance: $\prod s_i \in \{-1, +1\}$ (Heider-stable preferred)
- fraction_negative: $|s_i = -1|/k$ (frustration preferred)
- ...

Pruners that fire *during* DFS:

- BFS-distance pruner (always on, $\sim 10\times$ DFS savings)
- balance (only Heider-stable cycles)
- unbalanced (only frustrated cycles)
- ...



The headline result — a 4.4σ regime split

The right axiom is graph-conditional.

On Slashdot (5-seed, single-machine):

cycle pruner	mean AUC	vs no axiom
balance	0.8178 ± 0.008	-1.4σ (HURTS)
none	0.8359 ± 0.019	—
unbalanced	0.8562 ± 0.010	$+1.4\sigma$
unbalanced vs balance: $+0.0384$ AUC, 4.4σ, $p \ll 0.001$		

Why: Slashdot is an *adversarial* network — people post and downvote opinions they disagree with. Frustrated triads ($\prod s_i = -1$, e.g. “my friend’s friend is my enemy”) are the predictively informative cycles. Cooperative-network axioms *actively hurt* performance here.

On Bitcoin Alpha (cooperative trust): the regime is inverted. balance wins, unbalanced would hurt.

The right structural rule is graph-conditional, predictable from the fraction of balanced triangles (Heider 1946 → Cartwright-Harary 1956 → today).

Top- K outperforms full cycle enumeration!

Bitcoin Alpha, 5-seed, $m=128$ + balance pruner + $h=16$:

configuration	AUC	memory cut
Full cycle enumeration	0.9203	1×
Top- K + balance (mean)	0.9136 ± 0.020	$\sim 50\times$
Top- K + balance (best seed)	0.9329	$\sim 50\times$

Surprising finding: at the lucky-seed end of the distribution, the axiom-pruned top- K is *better* than exhaustive enumeration by +0.013 AUC — because full enumeration includes noisy cycles the model has to learn to ignore. The axiom *denoises* the cycle population.

The lesson: “brute force” isn’t a reliable maximum on information-rich datasets. Domain knowledge can beat exhaustion not just on *cost* but on *quality*.

Memory savings: $\sim 50\times$ on Bitcoin Alpha, $\sim 150\times$ on Slashdot, all without losing more than ~ 1 pp AUC compared to the impossible-to-run full enumeration.

Quaternion attention for signed graphs

Idea: standard scalar dot-product attention treats every embedding dimension symmetrically. For signed-graph data, we want the imaginary parts to encode *phase*.

Hamilton product real part as the score:

$$\text{score}(q, k) = \text{Re}(q \otimes k) = q_a k_a - q_b k_b - q_c k_c - q_d k_d$$

The (i, j, k) components contribute with negative sign — captures sign-rotation alignment that scalar attention misses.

5-seed result on Bitcoin Alpha at $m=16$ (same params: 61 860)

attention head	mean AUC
no attention	0.7819
dot product	0.7999
quaternion (real Hamilton)	0.8046

Quaternion beats dot consistently (+0.005, same param count). **But:** with axiom pruning (balance) on top, attention adds only +0.008 on top of the +0.043 that the axiom alone gave.

Reading: the axiom is the bigger lever: quaternion is a second-order improvement on the attention head

The sustainability angle — algorithm $>$ scale

Comparison: training campaign of 5 seeds $\times$ 3 datasets

approach	hardware	wall	energy	rel. CO ₂
“Buy more compute” SOTA	1 $\times$ A100, 40 GB	$\sim$ 24 h	$\sim$ 6 kWh	1.0 $\times$
Axiom-aware top-K	6-yr-old RTX 2070S, 8 GB	$\sim$ 2.5 h	$\sim$ 0.22 kWh	0.04 $\times$

25 $\times$ **less CO₂ for the same scientific result.**

On consumer hardware that already exists in millions of homes.

Where this matters:

- Reproducibility: any student with a gaming PC can replicate
- Deployment: edge inference, on-device sign prediction
- Climate: ML’s compute footprint is doubling every $\sim$ 3 months; algorithmic-efficiency wins compound
- *The P-graph methodology comes from sustainable chemistry* (1992, Friedler et al.) — using it for sustainable ML is meta-circular and elegant

Open direction — community-conditional axioms

Today's regime split is at the dataset level — one axiom per graph. But real graphs are *heterogeneous*: cooperative sub-communities next to conflict sub-communities, all in the same network.

Phase A (just landed today, late evening):

- 1 Detect communities via label propagation (or Louvain, signed Laplacian).
- 2 Compute per-community balance ratio.
- 3 Apply the right axiom *per community*.
- 4 Inter-community cycles get separate scoring.

Phase B / C (research-grade): replace the heuristic scorer with empirical **mutual information** between cycle features and edge labels:

$$I(\text{cycle}; \text{label}) = \sum_{c,y} p(c, y) \log \frac{p(c, y)}{p(c)p(y)}$$

Information-theoretic top- K : keep cycles that actually predict the label, regardless of their balance class.

Why this matters for AI students: this is the bridge from *combinatorial optimisation* (Fiedler P-graph 1992) to *information theory* (Shannon 1948) to *representation learning* (modern KAN). All three layers in one architecture.

Takeaways for AI students

- ① **Domain theory is a real lever.** A 1956 sociological theorem outperforms naive cycle aggregation by 4.4σ on Slashdot. *Read the literature outside ML.*
- ② **The right inductive bias beats parameter scaling.** 1.3 M params with axiom-aware pruning gets within 4 AUC points of 4 M-param transformer SOTA, with $25\times$ less compute.
- ③ **Brute force isn't a reliable upper bound.** Top- K with the right axiom *outperformed* full enumeration on Bitcoin Alpha. Less-but-better data $>$ more-but-noisy data.
- ④ **Geometry matters.** Quaternion attention (Hamilton product real part) beats dot-product attention at the same parameter count. The right algebra encodes the right structure.
- ⑤ **Sustainability is engineerable.** A $25\times$ CO_2 reduction came from algorithmic discipline, not new hardware. *Think before you compute.*

“When the search space is combinatorial and evaluations are expensive, axioms beat brute force.”

— the Friedler-P-graph posture, 1992, transposed to graph ML, 2026.